

HTML Forms — Notes

1. What is an HTML form?

An HTML form collects information from a user and can send it to a server for processing. A form may contain text fields, radio buttons, checkboxes, dropdowns, file inputs, and buttons.

```
<form action="register.php" method="post">
  <!-- Form controls go here -->
  <button type="submit">Register</button>
</form>
```

The `<form>` element defines the form. Each control needs a **name** if its value should be included in the submitted data.

2. Important `<form>` attributes

Attribute	Purpose	Example
action	URL that receives the form data	action="register.php"
method	HTTP method used for submission	method="post"
target	Where the response opens	target="_blank"
enctype	How submitted data is encoded; important for uploads	enctype="multipart/form-data"
autocomplete	Allows or restricts browser autofill	autocomplete="on"
novalidate	Skips built-in form validation on submission	novalidate

GET and POST

method="get"	method="post"
Adds form data to the URL as a query string	Sends form data in the request body
Useful for searches and filters	Common for registration, uploads, and data changes
URL can be bookmarked or shared	Submitted values do not appear in the URL
Unsuitable for passwords	Still requires HTTPS to protect data in transit

For example, submitting a GET form with `name="course"` and value `HTML` may produce:

```
search.php?course=HTML
```

3. Labels and text inputs

A `<label>` describes a control. Its `for` value should match the control's `id`. Clicking the label then focuses the control.

```
<label for="studentName">Student name:</label>
<input type="text" id="studentName" name="studentName">
```

Here, `id` connects the label to the field, while `name` identifies the value during submission.

Common input types

Type	Typical use	Example
text	Short text	Name
email	Email address	College email
password	Masked text entry	Password
tel	Telephone number	Mobile number
number	Numeric value	Age
date	Calendar date	Date of birth
time	Time of day	Appointment time
datetime-local	Local date and time	Meeting slot
range	Slider for a numeric value	Rating
color	Colour picker	Theme colour
file	File selection	Assignment upload
hidden	Value without a visible control	Form identifier

```
<input type="email" name="email">
<input type="tel" name="phone">
<input type="number" name="age" min="18" max="25">
<input type="time" name="meetingTime" min="09:00" max="17:00">
<input type="range" name="rating" min="1" max="10">
```

Remember: `type="tel"` can show a telephone keypad on mobile devices, but it does not automatically verify a phone-number format.

4. Radio buttons and checkboxes

Radio buttons: choose one option

Radio buttons in the same group must share a `name`.

```
<fieldset>
  <legend>Attendance mode</legend>

  <label>
    <input type="radio" name="mode" value="online" required>
    Online
  </label>

  <label>
    <input type="radio" name="mode" value="in-person">
    In person
  </label>
</fieldset>
```

Checkboxes: choose zero, one, or several options

```
<fieldset>
  <legend>Interested topics</legend>

  <label><input type="checkbox" name="topics" value="html"> HTML</label>
```

```

    <label><input type="checkbox" name="topics" value="css"> CSS</label>
    <label><input type="checkbox" name="topics" value="javascript">
JavaScript</label>
</fieldset>

```

A checkbox that is **not checked** is generally absent from the submitted data. For multiple selections, the server must be prepared to receive more than one value for `topics`.

5. Dropdowns, text areas, and grouped options

Use `<select>` for a list of choices and `<option>` for each choice.

```

<label for="course">Course:</label>
<select id="course" name="course" required>
  <option value="">Select a course</option>
  <option value="cse">Computer Science</option>
  <option value="ece">Electronics</option>
</select>

```

The value of the chosen `<option>` is submitted. In this example, selecting “Computer Science” submits `cse`.

Use `<textarea>` for longer, multi-line input:

```

<label for="message">Your message:</label>
<textarea id="message" name="message" rows="4" cols="40"></textarea>

```

For a long dropdown, `<optgroup>` can organise options:

```

<select name="workshop">
  <optgroup label="Web">
    <option value="html">HTML</option>
    <option value="css">CSS</option>
  </optgroup>
  <optgroup label="Programming">
    <option value="python">Python</option>
  </optgroup>
</select>

```

6. Input attributes and validation

Attribute	Effect
<code>required</code>	Field must have a value before submission
<code>minlength, maxlength</code>	Limits text length
<code>min, max</code>	Sets bounds for applicable inputs
<code>step</code>	Sets permitted numeric or time increments
<code>pattern</code>	Checks text against a regular expression
<code>placeholder</code>	Shows a temporary hint inside a field
<code>value</code>	Supplies an initial value
<code>readonly</code>	Value cannot be changed, but is usually submitted
<code>disabled</code>	Control cannot be used and is not submitted

```

<label for="password">Password:</label>
<input
  type="password"
  id="password"
  name="password"
  minlength="8"
  required
>

```

A phone-number format can be checked with pattern:

```

<label for="phone">10-digit phone number:</label>
<input
  type="tel"
  id="phone"
  name="phone"
  pattern="[0-9]{10}"
  title="Enter exactly 10 digits"
  required
>

```

The pattern checks the format entered in the browser. **The server must also validate submitted data**, because browser validation can be bypassed.

placeholder is not a label

A placeholder disappears when the user types and may be difficult to read. Give the field a visible `<label>` even when it has a placeholder.

7. Buttons

```

<button type="submit">Submit</button>
<button type="reset">Clear form</button>
<button type="button">Check availability</button>

```

- `submit` sends the form.
- `reset` restores controls to their initial values.
- `button` has no submission action by itself; JavaScript can give it an action.

Specify the button type explicitly so its behaviour is clear.

8. File uploads

For a file upload, use `method="post"` and `enctype="multipart/form-data"` on the form.

```

<form action="upload.php" method="post" enctype="multipart/form-data">
  <label for="assignment">Upload assignment (PDF):</label>
  <input
    type="file"
    id="assignment"
    name="assignment"
    accept=".pdf,application/pdf"
    required
  >

```

```
<button type="submit">Upload</button>
</form>
```

`accept` guides the file picker; it does not guarantee that the uploaded file is safe or actually a PDF. The server must check it.

9. Where does the form response appear?

The form's `target` attribute controls where the response is displayed.

Value	Result
<code>_self</code>	Current tab or frame; default
<code>_blank</code>	New tab or window
<code>_parent</code>	Parent frame
<code>_top</code>	Full browser tab, outside nested frames

An `iframe`'s name That named `iframe`

```
<form action="result.php" method="get" target="_blank">
  <input name="query">
  <button type="submit">Search</button>
</form>
```

10. Complete example: workshop registration

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Workshop Registration</title>
</head>
<body>
  <h1>Workshop Registration</h1>

  <form action="register.php" method="post">
    <fieldset>
      <legend>Student details</legend>

      <p>
        <label for="name">Full name:</label>
        <input type="text" id="name" name="name" required>
      </p>

      <p>
        <label for="email">Email:</label>
        <input type="email" id="email" name="email" required>
      </p>

      <p>
        <label for="phone">Phone number:</label>
        <input type="tel" id="phone" name="phone"
          pattern="[0-9]{10}" title="Enter 10 digits" required>
      </p>
    </fieldset>

    <fieldset>
```

```

<legend>Workshop details</legend>

<p>
  <label for="workshop">Select a workshop:</label>
  <select id="workshop" name="workshop" required>
    <option value="">Choose one</option>
    <option value="html">HTML Forms</option>
    <option value="css">CSS Design</option>
  </select>
</p>

<p>
  <label for="time">Preferred time:</label>
  <input type="time" id="time" name="time"
    min="09:00" max="17:00" required>
</p>

<p>
  <label for="comments">Questions or comments:</label><br>
  <textarea id="comments" name="comments" rows="4"></textarea>
</p>
</fieldset>

<p>
  <label>
    <input type="checkbox" name="agree" value="yes" required>
    I confirm that the information is correct.
  </label>
</p>

  <button type="submit">Register</button>
  <button type="reset">Reset</button>
</form>
</body>
</html>

```

Add these sections to the HTML Forms notes:

11. <fieldset> and <legend>

<fieldset> groups related form controls. <legend> gives that group a visible heading. They are useful when a form has sections or when several radio buttons or checkboxes answer one question.

```

<form action="register.php" method="post">
  <fieldset>
    <legend>Personal details</legend>

    <label for="fullName">Full name:</label>
    <input type="text" id="fullName" name="fullName" required>

    <label for="email">Email:</label>
    <input type="email" id="email" name="email" required>
  </fieldset>

  <fieldset>
    <legend>Attendance mode</legend>

```

```

    <label>
      <input type="radio" name="mode" value="online" required>
      Online
    </label>

    <label>
      <input type="radio" name="mode" value="campus">
      On campus
    </label>
  </fieldset>

  <button type="submit">Register</button>
</form>

```

Both attendance options use `name="mode"`, so the user can select only one. The `<legend>` tells the user what the group of choices is about.

12. `<datalist>`

`<datalist>` supplies suggestions for an `<input>` field. The user can choose a suggestion **or type a different value**.

Connect them by giving the `<input>` a `list` value that matches the `<datalist>` element's `id`.

```

<label for="city">City:</label>
<input type="text" id="city" name="city" list="cityOptions">

<datalist id="cityOptions">
  <option value="Chandigarh">
  <option value="Mohali">
  <option value="Panchkula">
  <option value="Patiala">
</datalist>

```

If the user selects “Mohali,” the form submits `city=Mohali`. If the user types “Delhi,” the form submits `city=Delhi`.

Element	Can the user type a value outside the listed options?
<code><datalist></code> linked to an <code><input></code>	Yes
<code><select></code>	No

Example: workshop registration with all three elements

```

<form action="register.php" method="post">
  <fieldset>
    <legend>Student details</legend>

    <label for="studentName">Name:</label>
    <input type="text" id="studentName" name="studentName" required>

    <label for="city">City:</label>
    <input type="text" id="city" name="city" list="cities" required>

```

```

    <datalist id="cities">
      <option value="Chandigarh">
      <option value="Mohali">
      <option value="Panchkula">
    </datalist>
  </fieldset>

  <fieldset>
    <legend>Workshop preference</legend>

    <label>
      <input type="radio" name="workshop" value="html" required>
      HTML
    </label>

    <label>
      <input type="radio" name="workshop" value="css">
      CSS
    </label>
  </fieldset>

  <button type="submit">Register</button>
</form>

```

Remember: <datalist> provides suggestions; it does not restrict the user to those suggestions.

13. Slider in an HTML form

To create a **slider**, use <input type="range">. It lets the user choose a number by moving a handle along a track.

```

<label for="rating">Rate the workshop:</label>
<input type="range" id="rating" name="rating"
  min="1" max="10" value="5">

```

Attribute	Purpose
min	Lowest allowed value
max	Highest allowed value
value	Initial position
step	Amount the value changes at each step

For example, this slider permits **0, 5, 10, ..., 100**:

```

<label for="volume">Volume:</label>
<input type="range" id="volume" name="volume"
  min="0" max="100" step="5" value="50">

```

A range input does not usually show its current number automatically. Use <output> if the user needs to see it:

```

<label for="volume">Volume:</label>
<input type="range" id="volume" name="volume"
  min="0" max="100" value="50"
  oninput="volumeValue.value = this.value">
<output id="volumeValue" for="volume">50</output>

```

Here, moving the slider updates the number displayed beside it.